ENTERPRISE ARCHITECTURE AND COMPONENTS

Practical Experiments – Explanations and Viva-Voce Answers

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How to Use This Document

This document explains all five experiments in simple language and provides answers to the viva-voce questions listed at the end of each experiment in the lab manual. The answers are written in a concise, student-friendly style so they can be spoken during a practical examination.

Tip: First understand the experiment explanation, then practise the viva answers aloud. Expand an answer with an example if the examiner asks a follow-up question.

EXPERIMENT 1

Enterprise Architecture Planning, Security and Governance

1. Experiment Explanation

Aim: To understand the planning, security and governance aspects of Enterprise Architecture (EA).

Enterprise Architecture is a structured way of aligning an organisation's business goals with its processes, information, applications and technology. It gives the organisation a common blueprint for how its different parts should work together.

Enterprise Architecture Planning

EA planning defines the organisation's present architecture, the architecture it wants to achieve and the steps needed to move from the present to the future state.

1. Understand business objectives: identify the organisation's goals and requirements.
2. Study the current architecture: document existing business processes, applications, data and infrastructure.
3. Perform gap analysis: compare the current state with the desired future state and identify what is missing.
4. Design the target architecture: define the future business, data, application and technology arrangements.
5. Prepare an implementation roadmap: decide the projects, priorities, resources and timelines.
6. Monitor and improve: review progress and update the architecture when business needs change.

Security in Enterprise Architecture

Security protects the organisation's systems, data and services from unauthorised access, misuse, damage and disruption. Important measures include:

- Identity and Access Management (IAM): verifies users and gives them only the access they need.
- Data protection: uses encryption, backups and secure handling of sensitive information.
- Network security: uses firewalls, secure connections and network monitoring.
- Risk assessment: identifies threats, vulnerabilities and their possible impact.
- Backup and recovery: helps restore systems and data after failures or attacks.
- Compliance and monitoring: checks whether security policies and legal requirements are followed.

Enterprise Architecture Governance

Governance is the system of rules, responsibilities and decision-making processes that ensures the architecture follows organisational goals and standards.

- Architecture principles and standards guide technology decisions.
- Roles and responsibilities make ownership clear.

- An Architecture Review Board (ARB) reviews proposed architecture changes and checks whether they follow standards.
- Compliance checks identify deviations from approved policies.
- Change management controls changes to architecture.
- Performance monitoring checks whether the architecture is meeting its objectives.

Benefits include better alignment with business goals, reduced duplication, improved security, better decision-making and controlled technology costs. Common challenges include resistance to change, lack of skilled people, limited budget and difficulty integrating old systems.

2. Viva-Voce Questions and Answers

Q1. What is Enterprise Architecture (EA)?

Answer: Enterprise Architecture is a structured blueprint that explains how an organisation's business processes, data, applications and technology work together to achieve its goals.

Q2. Why is Enterprise Architecture planning important?

Answer: It helps an organisation understand its current systems, define a future direction, identify gaps and prepare a roadmap. It also reduces duplication and improves the use of resources.

Q3. What is the difference between current state and target state architecture?

Answer: Current state describes the systems and processes that exist today. Target state describes the desired future systems and processes that the organisation plans to achieve.

Q4. What is gap analysis in EA?

Answer: Gap analysis compares the current architecture with the target architecture. It identifies missing capabilities, outdated systems and changes required to reach the target state.

Q5. Why is security important in Enterprise Architecture?

Answer: Security protects organisational data, applications and infrastructure from unauthorised access, cyberattacks, data loss and service disruption.

Q6. What is Identity and Access Management (IAM)?

Answer: IAM is a framework used to identify and authenticate users and control what resources they are allowed to access.

Q7. What is Enterprise Architecture governance?

Answer: It is the set of policies, standards, roles and review processes that ensures architecture decisions support business goals and follow agreed rules.

Q8. What is the role of an Architecture Review Board (ARB)?

Answer: The ARB reviews proposed architecture designs and changes. It checks whether they follow organisational standards, principles, security requirements and long-term plans.

Q9. State four benefits of Enterprise Architecture governance.

Answer: It improves consistency, aligns technology with business goals, reduces unnecessary duplication and helps control risk and technology costs.

Q10. What are three challenges in implementing Enterprise Architecture?

Answer: Common challenges are resistance to organisational change, lack of skilled resources and limited budget. Integrating legacy systems can also be difficult.

EXPERIMENT 2

Integrating Cloud, IoT, AI and Blockchain in Enterprise Architecture

1. Experiment Explanation

Aim: To study how modern technologies such as Cloud Computing, Internet of Things (IoT), Artificial Intelligence (AI) and Blockchain can be integrated into Enterprise Architecture.

Enterprise Architecture connects business requirements with four main architecture domains:

- Business architecture: describes business goals, functions, processes and organisational roles.
- Data architecture: describes what data is required, where it is stored and how it flows.
- Application architecture: describes software applications and how they interact.
- Technology architecture: describes infrastructure such as servers, networks, cloud platforms and devices.

Role of Modern Technologies

- Cloud computing provides computing resources such as servers, storage, databases and software over the internet. It supports scalability, flexibility and reduced infrastructure maintenance.
- IoT connects physical devices and sensors to networks so they can collect and exchange data. Examples include smart meters, health-monitoring devices and connected machines.
- AI analyses data, recognises patterns, makes predictions and supports automated or informed decisions. Examples include demand forecasting and anomaly detection.
- Blockchain is a distributed digital ledger in which records are stored in linked blocks. It can provide tamper-evident records, traceability and shared verification among participants.

Example: Smart College

A smart-college system can use IoT sensors to collect attendance or energy-use data. The data can be sent to a cloud platform for storage. AI can analyse the data to identify patterns, such as unusual energy consumption. Blockchain can be used where a tamper-evident record is required, for example, for verifying certificates. The business, data, application and technology architecture domains define how these parts connect.

Security and governance apply across all layers. Access control, encryption, monitoring, data-quality rules and clear ownership are needed. Benefits include improved efficiency, scalability, automation and transparency. Challenges include integration complexity, privacy concerns, security risks, cost and the need for skilled staff.

2. Viva-Voce Questions and Answers

Q1. What is Enterprise Architecture?

Answer: Enterprise Architecture is a blueprint that aligns an organisation's business processes, data, applications and technology with its goals.

Q2. What are the four main domains of Enterprise Architecture?

Answer: The four domains are Business Architecture, Data Architecture, Application Architecture and Technology Architecture.

Q3. What is the role of cloud computing in Enterprise Architecture?

Answer: Cloud computing provides scalable computing, storage, databases and software services over the internet. It helps organisations increase or decrease resources as needed.

Q4. What is IoT? Give examples.

Answer: The Internet of Things is a network of physical devices containing sensors or software that collect and exchange data. Examples include smart watches, smart meters, connected vehicles and industrial sensors.

Q5. How does AI help in Enterprise Architecture?

Answer: AI analyses large amounts of data, identifies patterns, predicts outcomes and supports decision-making. It can also automate tasks such as anomaly detection.

Q6. What is Blockchain?

Answer: Blockchain is a distributed digital ledger that stores records in linked blocks. Once records are confirmed and added, changing them is difficult, which supports traceability and trust.

Q7. How can Blockchain be used for certificate verification?

Answer: A certificate's digital details or its hash can be recorded on a blockchain. A verifier can compare the certificate with the recorded information to check whether it is genuine and unchanged.

Q8. What is the role of Data Architecture?

Answer: Data Architecture defines the organisation's data, its structure, storage, movement, integration, quality and governance.

Q9. Why is an architecture diagram useful?

Answer: It visually shows the components of a system and how they connect. It helps stakeholders understand data flow, technology dependencies and integration points.

Q10. Why are security and governance required across all architecture layers?

Answer: Every layer handles important resources and information. Security protects them, while governance ensures consistent standards, accountability, compliance and controlled changes.

EXPERIMENT 3

Security Architecture Using SABSA

1. Experiment Explanation

Aim: To understand how to design a business-driven security architecture using the SABSA framework for domains such as Finance, Defence or Agriculture.

SABSA stands for Sherwood Applied Business Security Architecture. It is a risk-driven framework that starts with business requirements and develops security architecture to protect the organisation's important assets. Security controls are selected according to business risks rather than added without a clear purpose.

Six SABSA Layers / Views

7. Contextual: identifies business requirements, goals, assets, risks and the reasons security is needed.
8. Conceptual: defines the high-level security strategy and principles.
9. Logical: describes logical security services and structures, such as authentication and access control.
10. Physical: identifies the technologies and infrastructure used to implement the logical design.
11. Component: specifies detailed products, configurations and components.
12. Operational: covers day-to-day operation, monitoring, maintenance and incident response.

Important Security Concepts

- Asset: something valuable that must be protected, such as customer records, payment systems or farm data.
- Threat: something that can cause harm, such as malware, fraud, theft or a natural event.
- Vulnerability: a weakness that a threat can exploit.
- Risk: the possibility of loss or harm when a threat exploits a vulnerability, considered in terms of likelihood and impact.
- Least privilege: users receive only the minimum permissions needed for their work.

- Encryption: converts readable information into an encoded form so unauthorised people cannot easily understand it.
- Monitoring and incident response: detect suspicious activity, investigate it, contain the incident and recover services.

The security controls differ by domain. Finance may prioritise transaction integrity, customer-data confidentiality, fraud prevention and strong authentication. Agriculture may prioritise protection of sensor networks, farm-management data, connected equipment and availability of monitoring systems.

2. Viva-Voce Questions and Answers

Q1. Why is SABSA called a business-driven security architecture framework?

Answer: Because it begins with business objectives, assets and risks. Security requirements and controls are designed to support the organisation's business needs.

Q2. What is the difference between a threat, vulnerability and risk?

Answer: A threat is a possible source of harm. A vulnerability is a weakness that can be exploited. Risk is the possibility and impact of harm when a threat exploits a vulnerability.

Q3. Name the six layers of the SABSA framework.

Answer: The six layers are Contextual, Conceptual, Logical, Physical, Component and Operational.

Q4. What is the principle of least privilege?

Answer: It means giving each user or system only the minimum access permissions required to perform its assigned tasks.

Q5. Why is encryption important in security architecture?

Answer: Encryption protects data by converting it into an unreadable form for unauthorised users. It helps protect data during storage and transmission.

Q6. Why are monitoring and incident response important?

Answer: Monitoring helps detect suspicious activities and security events. Incident response provides a planned way to investigate, contain, remove and recover from security incidents.

Q7. How does security architecture differ between Finance and Agriculture?

Answer: Finance focuses strongly on transaction security, fraud prevention, customer privacy and regulatory requirements. Agriculture may focus more on IoT sensors, connected equipment, farm data and maintaining the availability of monitoring systems.

EXPERIMENT 4

Enterprise Information Architecture: Component and Operational Models

1. Experiment Explanation

Aim: To study the components of Enterprise Information Architecture (EIA) and understand the operational model used to manage organisational information.

Enterprise Information Architecture is the structure and set of practices used to manage an organisation's information. It explains how data is collected, integrated, stored, governed, protected and delivered to users.

Main Components of EIA

13. Source systems: generate data, for example, ERP, CRM, websites and operational applications.

14. Integration layer: moves data between systems using APIs, messaging or data-integration processes.
15. Operational database: stores current data needed for day-to-day transactions.
16. Data warehouse and analytics: stores integrated historical data for reporting and analysis.
17. Metadata and master data: metadata describes data; master data maintains consistent core entities such as customers or products.
18. Security and access control: protects information and ensures only authorised access.
19. Data quality and governance: defines ownership, standards, validation and rules for reliable data.
20. Presentation and reporting: provides dashboards, reports and other views for users.

Operational Model and Roles

The operational model explains who manages the information systems and how information-related activities are performed.

- Data owner: accountable for a data domain and its appropriate use.
- Data steward: helps maintain data definitions, quality and consistency.
- Database administrator: manages database performance, availability, backup and access.
- Integration team: manages data movement and connections between systems.
- Security officer: oversees information security and access policies.
- Business user: uses information for daily work and decisions.
- Management or governance committee: sets policies, priorities and accountability.

Important Terms

- Operational database: supports current, routine transactions such as placing an order or updating a customer record.
- Data warehouse: stores integrated historical data, mainly for reporting, analysis and decision support.
- Master Data Management (MDM): manages important shared entities consistently across systems.
- Metadata: data about data, such as its meaning, format, source, owner and update frequency.
- API: Application Programming Interface; a defined way for software systems to communicate.

A typical flow is: source systems → integration layer → operational databases and/or data warehouse → reporting and business users. Security, data quality and governance apply throughout the flow.

2. Viva-Voce Questions and Answers

Q1. What is Enterprise Information Architecture (EIA)?

Answer: EIA is the structure and practices used to organise, integrate, store, manage, protect and deliver an organisation's information.

Q2. What is an EIA component model?

Answer: It is a representation of the major parts involved in managing information, such as source systems, integration, databases, data warehouses, metadata, security, governance and reporting.

Q3. What is an operational model?

Answer: An operational model explains the roles, responsibilities, processes and activities required to operate and maintain the information architecture.

Q4. Differentiate between an operational database and a data warehouse.

Answer: An operational database supports current day-to-day transactions and frequent updates. A data warehouse stores integrated historical data for analysis, reporting and decision-making.

Q5. What is Master Data Management (MDM)?

Answer: MDM is the process of creating and maintaining consistent, reliable core business data, such as customer, supplier and product information, across different systems.

Q6. What is metadata?

Answer: Metadata is data about data. It describes information such as a dataset's name, meaning, format, source, owner and update frequency.

Q7. Who is a data steward?

Answer: A data steward is responsible for helping maintain data quality, consistency, definitions and proper use within a particular data area.

Q8. Why is data quality important?

Answer: Good data quality makes information accurate, complete, consistent and reliable. It improves reporting and helps people make correct decisions.

Q9. What is the role of an API layer in EIA?

Answer: The API layer allows different applications and systems to communicate through defined interfaces. It supports integration and controlled data exchange.

Q10. Why is security required across all EIA components?

Answer: Data passes through many systems and layers. Security controls protect it from unauthorised access, alteration, disclosure and loss throughout its lifecycle.

EXPERIMENT 5

Enterprise Architecture Management Tools

1. Experiment Explanation

Aim: To study and compare Enterprise Architecture Management (EAM) tools such as SAP LeanIX, ADOIT, UPMX and Avolution ABACUS.

Enterprise Architecture Management tools help organisations document, analyse, visualise and manage their architecture. They provide a central place to store information about business capabilities, applications, data, technology, relationships, risks and transformation plans.

Purpose and Common Features

- Architecture repository: central storage for architecture information and diagrams.
- Application portfolio management: records applications, owners, costs, technologies and lifecycle status.
- Visual modelling: creates diagrams and views of business and technology relationships.
- Impact analysis: helps identify which systems or processes may be affected by a proposed change.
- Roadmap planning: supports planning and tracking architecture transformation.
- Governance: helps check standards, ownership, risks and compliance.
- Integration: connects with other enterprise systems and data sources.

Tools Mentioned in the Experiment

- SAP LeanIX: a commercial, cloud-based enterprise architecture management platform used for application portfolio management, technology risk and transformation planning.
- ADOIT: an enterprise architecture management tool from BOC Group. The lab manual refers to its Community Edition; availability and capabilities can differ from commercial offerings.

- UPMX: an architecture and governance-related solution associated with the ServiceNow ecosystem, used to support architecture information and workflows.
- Avolution ABACUS: a commercial enterprise architecture and business process modelling platform with modelling, analysis and repository capabilities.

How to Select an EAM Tool

Selection depends on organisational needs, ease of use, modelling capabilities, integration with existing systems, scalability, reporting, security, licensing cost, vendor support and data portability.

Important distinction: open-source software makes its source code available under an open-source licence. A free edition or community edition may be free to use but can still be proprietary and have feature or usage limits. Therefore, free does not automatically mean open source.

2. Viva-Voce Questions and Answers

Q1. What is Enterprise Architecture Management (EAM)?

Answer: EAM is the practice of documenting, analysing, governing and improving an organisation's enterprise architecture so that technology and business remain aligned.

Q2. What is an architecture repository?

Answer: It is a central storage location for architecture information, such as applications, business capabilities, data, technologies, diagrams, owners and relationships.

Q3. What is the difference between open-source software and a free edition?

Answer: Open-source software makes its source code available under an open-source licence, allowing use and modification according to that licence. A free edition may be free to use but can be proprietary and may have limitations.

Q4. What is SAP LeanIX used for?

Answer: SAP LeanIX is used for enterprise architecture management, especially application portfolio management, technology-risk visibility and transformation planning.

Q5. What is ADOIT Community Edition?

Answer: ADOIT is an enterprise architecture management tool from BOC Group. Its Community Edition is an entry-level offering for exploring architecture modelling and management; its exact features may differ from paid editions.

Q6. What is UPMX and how is it related to ServiceNow?

Answer: UPMX is presented in the experiment as an architecture and governance-related solution associated with the ServiceNow ecosystem. It supports architecture information and workflow-related activities.

Q7. What is a configurable metamodel?

Answer: A metamodel defines the types of architecture objects, their properties and the relationships between them. A configurable metamodel allows an organisation to adapt these definitions to its own requirements.

Q8. Why are integrations important in EAM tools?

Answer: Integrations allow an EAM tool to exchange information with systems such as IT service management, project management, cloud platforms and application databases. This reduces manual data entry and keeps information more consistent.

Q9. State four criteria for selecting an EAM tool.

Answer: Four criteria are required modelling capabilities, ease of use, integration with existing systems and total cost/licensing. Scalability, security, vendor support and reporting are also important.

Q10. Why are licensing and data portability important?

Answer: Licensing determines the cost and permitted use of the tool. Data portability helps an organisation export or transfer its architecture information if it changes tools or vendors.

QUICK REVISION – IMPORTANT DIFFERENCES

1. Current State vs Target State

Current State	Target State
What exists in the organisation now.	What the organisation wants to achieve in the future.

2. Threat vs Vulnerability vs Risk

Term	Meaning
Threat	A possible source of harm.
Vulnerability	A weakness that can be exploited.
Risk	The likelihood and impact of harm when a threat exploits a vulnerability.

3. Operational Database vs Data Warehouse

Operational Database	Data Warehouse
Supports current transactions and frequent updates.	Stores integrated historical data for analysis and reporting.

4. Open Source vs Free Edition

Open Source	Free Edition
Source code is available under an open-source licence.	Can be free to use but may be proprietary and feature-limited.

5. SABSA Layers – Remember in Order

Contextual → Conceptual → Logical → Physical → Component → Operational

6. Four EA Domains – Remember

Business → Data → Application → Technology

7. Last-Minute Viva Tips

- Start with a one-sentence definition.
- Explain the purpose in simple words.
- Give one relevant example if asked.
- For comparisons, state the main difference directly.
- If you do not know a detail, explain the related concept honestly rather than guessing.